

Offline causal targeting case study: response at 2%

DECISION

For a hypothetical margin of 100 and cost of 0.50 per assigned target, select the response ranking and target the top 2% of eligible records. Response is the benchmark champion among the evaluated fixed linear baselines.

FROZEN OPERATING POINT

2%

Selected on development validation and locked before final access.

FINAL NET VALUE

+5,110

Conditional row-level paired-bootstrap
95% CI: +1,816 to +8,172 hypothetical units.

CHALLENGER VERDICT

Retain response

T minus response -1,778; superiority was not demonstrated.

CHALLENGE

The task was not simply to predict conversion. It was to choose a targeting policy and budget when every assigned target has a cost, while avoiding selection on the final evaluation sample.

APPROACH

Fit response, S-learner, and T-learner rankings on 1.6 million development-training records. Select the economic decision on 400,000 validation records, freeze it, then evaluate once on 400,000 disjoint final records.

CANDIDATE LOGIC

RESPONSE

benchmark champion

T-LEARNER

retained challenger

TARGET NOBODY

explicit no-action baseline

Economic scope was manually fixed as response, T-learner, and target nobody; S-learner was not excluded by an executable screen. At 20%, response 1.055, T 0.890, and S 0.751 HT-style IPW conversions per 1,000 eligible records. This is not a global inferiority claim.

OUTCOME

The frozen response policy produced 5109.98 incremental net value versus target nobody. Its conditional row-level paired-bootstrap interval was positive. T-learner did not pass the promotion rule, so the benchmark was retained.

WHAT THIS DEMONSTRATES

An offline causal targeting and policy-evaluation case study with explicit no-action economics, validation-only selection, frozen decisions, paired uncertainty, and a bounded recommendation.

FUTURE DEPLOYMENT WORK

Not built here: serialized encoder/model bundle, feature contract, deterministic top-k tie-breaking, batch-scoring entry point, monitoring, and prospective online validation.

BOUNDARIES

Rows are records, not verified users. The primary estimator is empirical-marginal-propensity IPW (legacy HT-style). CIs condition on rows, fitted models, rankings, and cutoff; they omit refit, selection, and latent clusters. Valid cluster IDs are unavailable; do not invent pseudo-clusters.